

1.METHODOLOGY USED FOR EXPENSE SHARING:

we use “**equal contribution method**”, like whenever a participant pays for a group expense, the total amount is divided equally among the number of participants involved in that expense.

Procedure:

1. The payer records the expense amount.
2. The participants involved in that expense are specified.
3. The total expense is divided equally among all the participants.
4. Each participant's balance is reduced by their share, whereas the payer's balance is increased by the total amount paid.
5. After all expenses are entered, settlements are calculated to determine who owes whom.

SAMPLE OUTPUT:

```
PS C:\Users\kavya sri CM> python sharing_app.py
enter the names of friends, seperated by commasAlice, bob, carol
enter the name of the person who paid (of 'done' to finish):Alice
enter the amount paid:1000
enter the names of participants for this expense, seperated by commasAlice, bob, carol
enter the name of the person who paid (of 'done' to finish):bob
enter the amount paid:1100
enter the names of participants for this expense, seperated by commasAlice, bob
enter the name of the person who paid (of 'done' to finish):carol
enter the amount paid:900
enter the names of participants for this expense, seperated by commasAlice, carol
enter the name of the person who paid (of 'done' to finish):done

Final settlement
Alice owes carol: Rs.116.67
Alice owes bob: Rs.216.67
```

2.DATASET DETAILS AND PROCESSING STEPS:

DATASET DETAILS:

Dataset attributes used in this project are “Payer”, “Amount”, “Participants”, “Balance”.

The description for the attributes are “name of the person who paid the expense”, “total amount paid”, “list of individuals sharing the expense”, “running net balance for each participant” respectively .

PROCESSING STEPS:

1. Input Cleaning:

Extra spaces are removed by strip().

friends = [friend.strip() for friend in friends]

2. Data Validation:

The system ensures that expense amounts are numeric values, Participants names exist in the registered friend list, The participant list is not empty.

3. Settlement preparation:

Positive balances represent creditors, while negative balances represent debtors.

3.KEY INSIGHTS AND HANDLING OF SPECIAL CASES:

Key Insights:

1. Simplified debt settlement
2. Efficient balance tracking
3. Scalability

Handling Special Cases:

1. Refunds

Refunds occur when money previously paid is returned. A refund can be treated as a negative expense. This ensures balances remain accurate

2. Invalid participants

If an unregistered participant is entered. The transaction should be rejected until valid names are provided.

3. Zero or Negative Amounts:

The system should prevent entries because they do not represent valid expenses.